OS Jackfruit — Multi-Container Runtime

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Screenshot Evidence — All 8 Required Demonstrations

Screenshot 1 — Multi-Container Supervision

Demonstrates: Two or more containers running under one supervisor process simultaneously.

Two containers (alpha pid=3619, beta pid=3624) both in running state under one supervisor. Started with engine start commands.

```
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ sudo dmesg | tail -5
[11125.413426] [container_monitor] SOFT LIMIT container=memtest pid=6664 rss=17170432 limit=10485760
[11126.440267] [container_monitor] HARD LIMIT container=memtest pid=6664 rss=25559040 limit=20971520
[11126.443905] [container_monitor] Unregister request container=memtest pid=6664
[15782.003914] [container_monitor] Module unloaded.
[16225.798605] [container_monitor] Module loaded. Device: /dev/container_monitor
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ ls -l /dev/container_monitor
crw------- 1 root root 235, 0 Apr 16 08:48 /dev/container_monitor
```

```
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ sudo ./engine supervisor ./rootfs-base
[supervisor] Ready. rootfs=./rootfs-base socket=/tmp/mini_runtime.sock
```

Screenshot 2 — Metadata Tracking

Demonstrates: Output of the ps command showing tracked container metadata (ID, PID, STATE, STARTED, LOG).

engine ps shows ID, PID, STATE, start time, and log file path for every tracked container.

```
cp: cannot create regular file './rootfs-alpha/': Not a directory
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ cp cpu_hog ../rootfs-alpha/
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ cp memory_hog ../rootfs-alpha/
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ cp cpu_hog ../rootfs-beta/
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ cp memory_hog ../rootfs-beta/
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ sudo ./engine start alpha ../rootfs-alpha /cpu_hog --soft-mib 48 --hard-mib 80
[sudo] password for sanchita:
^[[ASorry, try again.
[sudo] password for sanchita:
sudo: 1 incorrect password attempt
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ sudo ./engine start alpha ../rootfs-alpha /cpu_hog --soft-mib 48 --hard-mib 80
[sudo] password for sanchita:
started container alpha pid=8017
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ sudo ./engine start beta ../rootfs-beta /cpu_hog --soft-mib 64 --hard-mib 96
started container beta pid=8022
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ sudo ./engine ps
ID              PID     STATE     STARTED             LOG
beta            8022    running   09:02:23            logs/beta.log
alpha           8017    exited    09:02:14            logs/alpha.log
```

Screenshot 3 — Bounded-Buffer Logging

Demonstrates: Log file contents captured through the logging pipeline; evidence of producer/consumer activity.

```
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ sudo ./engine logs alpha
log: logs/alpha.log
cpu_hog alive elapsed=1 accumulator=2405451544717435047
cpu_hog alive elapsed=2 accumulator=2892213073434298327
cpu_hog alive elapsed=3 accumulator=6099240897903465387
cpu_hog alive elapsed=4 accumulator=17439775449186011368
cpu_hog alive elapsed=5 accumulator=15644117090703945808
cpu_hog alive elapsed=6 accumulator=11993325180609575654
cpu_hog alive elapsed=7 accumulator=10043066098913978240
cpu_hog alive elapsed=8 accumulator=1070575132873723605
cpu_hog alive elapsed=9 accumulator=3746510666945630435
cpu_hog done duration=10 accumulator=7708748353372337221
cpu_hog alive elapsed=1 accumulator=1385343786926085769
cpu_hog alive elapsed=2 accumulator=105796417938851238
cpu_hog alive elapsed=3 accumulator=14879673021820618754
cpu_hog alive elapsed=4 accumulator=15179030408504208342
cpu_hog alive elapsed=5 accumulator=534136573993055939
cpu_hog alive elapsed=6 accumulator=10163287918730053001
cpu_hog alive elapsed=7 accumulator=10747018871032947165
cpu_hog alive elapsed=8 accumulator=7322559666170659659
cpu_hog alive elapsed=9 accumulator=4333289991160498677
cpu_hog done duration=10 accumulator=237693329428382147
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ cat logs/alpha.log
cpu_hog alive elapsed=1 accumulator=2405451544717435047
cpu_hog alive elapsed=2 accumulator=2892213073434298327
cpu_hog alive elapsed=3 accumulator=6099240897903465387
cpu_hog alive elapsed=4 accumulator=17439775449186011368
cpu_hog alive elapsed=5 accumulator=15644117090703945808
cpu_hog alive elapsed=6 accumulator=11993325180609575654
cpu_hog alive elapsed=7 accumulator=10043066098913978240
cpu_hog alive elapsed=8 accumulator=1070575132873723605
cpu_hog alive elapsed=9 accumulator=3746510666945630435
cpu_hog done duration=10 accumulator=7708748353372337221
cpu_hog alive elapsed=1 accumulator=1385343786926085769
cpu_hog alive elapsed=2 accumulator=105796417938851238
cpu_hog alive elapsed=3 accumulator=14879673021820618754
cpu_hog alive elapsed=4 accumulator=15179030408504208342
cpu_hog alive elapsed=5 accumulator=534136573993055939
cpu_hog alive elapsed=6 accumulator=10163287918730053001
cpu_hog alive elapsed=7 accumulator=10747018871032947165
cpu_hog alive elapsed=8 accumulator=7322559666170659659
cpu_hog alive elapsed=9 accumulator=4333289991160498677
cpu_hog done duration=10 accumulator=237693329428382147
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$
```

engine logs alpha returns log data routed through the bounded-buffer pipeline. cat logs/alpha.log confirms persistence to disk. Each cpu_hog line was written by the consumer thread after the producer read it from the container pipe

Screenshot 4 — CLI and IPC

Demonstrates: CLI command issued to supervisor over UNIX domain socket, supervisor responds.

```
cpu_hog done duration=10 accumulator=237693329428382147
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ sudo ./engine stop alpha
sent SIGTERM to container alpha (pid=8017)
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ sudo ./engine stop beta
sent SIGTERM to container beta (pid=8022)
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ sudo ./engine ps
ID                  PID       STATE       STARTED                LOG
beta                8022      stopped     09:02:23               logs/beta.log
alpha               8017      stopped     09:02:14               logs/alpha.log

sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$
```

```
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ sudo ./engine start memtest ../rootfs-alpha /memory_hog --soft-mib 10 --hard-mib 20
started container memtest pid=8053
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ sudo dmesg -w
[    0.000000] Booting Linux on physical CPU 0x0000000000 [0x610f0000]
[    0.000000] Linux version 6.8.0-110-generic (buildd@bos03-arm64-019) (aarch64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0, GNU ld (GNU Binutils f
or Ubuntu) 2.42) #110-Ubuntu SMP PREEMPT_DYNAMIC Thu Mar 19 17:16:23 UTC 2026 (Ubuntu 6.8.0-110.110-generic 6.8.12)
[    0.000000] KASLR enabled
[    0.000000] efi: EFI v2.7 by EDK II
[    0.000000] efi: SMBIOS 3.0=0x13f810000 MEMATTR=0x13e803018 ACPI 2.0=0x13c150018 MOKvar=0x13c060000 INITRD=0x13c483a18 RNG=0x13c15e698 MEMRESERVE=0x13c483618
[    0.000000] random: crng init done
[    0.000000] secureboot: Secure boot disabled
[    0.000000] ACPI: Early table checksum verification disabled
[    0.000000] ACPI: RSDP 0x000000013C150018 000024 (v02 BOCHS )
[    0.000000] ACPI: XSDT 0x000000013C15FE98 00006C (v01 BOCHS  BXPC     00000001      01000013)
[    0.000000] ACPI: FACP 0x000000013C15FA98 000114 (v06 BOCHS  BXPC     00000001 BXPC 00000001)
[    0.000000] ACPI: DSDT 0x000000013C157518 0014A2 (v02 BOCHS  BXPC     00000001 BXPC 00000001)
[    0.000000] ACPI: APIC 0x000000013C15FC18 00019C (v04 BOCHS  BXPC     00000001 BXPC 00000001)
[    0.000000] ACPI: PPTT 0x000000013C15D898 000088 (v02 BOCHS  BXPC     00000001 BXPC 00000001)
[    0.000000] ACPI: GTDT 0x000000013C15E818 000068 (v03 BOCHS  BXPC     00000001 BXPC 00000001)
[    0.000000] ACPI: MCFG 0x000000013C15E918 00003C (v01 BOCHS  BXPC     00000001 BXPC 00000001)
[    0.000000] ACPI: SPCR 0x000000013C15FF98 000050 (v02 BOCHS  BXPC     00000001 BXPC 00000001)
[    0.000000] ACPI: DBG2 0x000000013C15E418 000057 (v00 BOCHS  BXPC     00000001 BXPC 00000001)
[    0.000000] ACPI: IORT 0x000000013C15E718 000080 (v03 BOCHS  BXPC     00000001 BXPC 00000001)
[    0.000000] ACPI: BGRT 0x000000013C15E498 000038 (v01 INTEL  EDK2     00000002      01000013)
[    0.000000] ACPI: SPCR: console: pl011,mmio32,0x9000000,9600
[    0.000000] NUMA: Failed to initialise from firmware
[    0.000000] NUMA: Faking a node at [mem 0x0000000040000000-0x000000013fffffff]
[    0.000000] NUMA: NODE_DATA [mem 0x13f64b5c0-0x13f650fff]
[    0.000000] Zone ranges:
[    0.000000]   DMA      [mem 0x0000000040000000-0x00000000ffffffff]
[    0.000000]   DMA32    empty
[    0.000000]   Normal   [mem 0x0000000100000000-0x000000013fffffff]
[    0.000000]   Device   empty
[    0.000000] Movable zone start for each node
[    0.000000] Early memory node ranges
[    0.000000]   node   0: [mem 0x0000000040000000-0x000000013c05ffff]
[    0.000000]   node   0: [mem 0x000000013c060000-0x000000013c06ffff]
[    0.000000]   node   0: [mem 0x000000013c070000-0x000000013c15ffff]
[    0.000000]   node   0: [mem 0x000000013c160000-0x000000013c47ffff]
[    0.000000]   node   0: [mem 0x000000013c480000-0x000000013f75ffff]
[    0.000000]   node   0: [mem 0x000000013f760000-0x000000013f7effff]
[    0.000000]   node   0: [mem 0x000000013f7f0000-0x000000013f7fffff]
[    0.000000]   node   0: [mem 0x000000013f800000-0x000000013f91ffff]
```

engine logs alpha sends a control_request_t over /tmp/mini_runtime.sock. Supervisor responds with "log: logs/alpha.log" and streams log content back — demonstrating the second IPC mechanism (distinct from logging pipes).

Screenshot 6 — Hard-Limit Enforcement

emonstrates: dmesg showing container killed after exceeding hard limit; kernel module removes the entry.

```
[17243.569536] [container_monitor] HARD LIMIT container=memtest pid=8046 rss=25559040 limit=20971520
[17243.573205] [container_monitor] Unregister request container=memtest pid=8046
[17277.827473] [container_monitor] Registering container=memtest pid=8053 soft=10485760 hard=20971520
[17279.407708] [container_monitor] SOFT LIMIT container=memtest pid=8053 rss=17170432 limit=10485760
[17280.428237] [container_monitor] HARD LIMIT container=memtest pid=8053 rss=25559040 limit=20971520
[17280.434016] [container_monitor] Unregister request container=memtest pid=8053
```

HARD LIMIT: container=memtest pid=3731 rss=25559040 limit=20971520 — SIGKILL sent by kernel module. Entry removed from monitored list. Supervisor metadata reflects killed state.

Screenshot 7 — Scheduling Experiment

Demonstrates: Two CPU-bound containers running at different nice values (-10 vs +10); observable scheduling

```
[sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ sudo ./engine logs highpri
 log: logs/highpri.log
 cpu_hog alive elapsed=1 accumulator=3417919832213694393
 cpu_hog alive elapsed=2 accumulator=16323725293504839762
 cpu_hog alive elapsed=3 accumulator=11613303308020667235
 cpu_hog alive elapsed=4 accumulator=11395399076287400298
 cpu_hog alive elapsed=5 accumulator=9992496982050188482
 cpu_hog alive elapsed=6 accumulator=5946128696837926236
 cpu_hog alive elapsed=7 accumulator=985482064164096726
 cpu_hog alive elapsed=8 accumulator=2919979404822979494
 cpu_hog alive elapsed=9 accumulator=132790945250691571
 cpu_hog done duration=10 accumulator=14301234740551259006
 cpu_hog alive elapsed=1 accumulator=17255367028732266569
 cpu_hog alive elapsed=2 accumulator=2309567426952111513
 cpu_hog alive elapsed=3 accumulator=4162459995508618924
 cpu_hog alive elapsed=4 accumulator=9157256561877222704
 cpu_hog alive elapsed=5 accumulator=93014102485621751
 cpu_hog alive elapsed=6 accumulator=245125454552747831
 cpu_hog alive elapsed=7 accumulator=16498578877710207672
 cpu_hog alive elapsed=8 accumulator=2035947424895144968
 cpu_hog alive elapsed=9 accumulator=12290428092158454440
 cpu_hog done duration=10 accumulator=16572297445218807277
 cpu_hog alive elapsed=1 accumulator=6009918915396607481
 cpu_hog alive elapsed=2 accumulator=6395402136853361816
 cpu_hog alive elapsed=3 accumulator=13341234159811186745
 cpu_hog alive elapsed=4 accumulator=13569079251853436659
 cpu_hog alive elapsed=5 accumulator=2083289227495601725
 cpu_hog alive elapsed=6 accumulator=8477207306176588985
 cpu_hog alive elapsed=7 accumulator=4187312651870135042
 cpu_hog alive elapsed=8 accumulator=2439043021246657561
 cpu_hog alive elapsed=9 accumulator=1028708360045800045
 cpu_hog done duration=10 accumulator=2916365152746435765
```

highpri (nice=-10) and lowpri (nice=10) both run cpu_hog for 10 seconds. highpri accumulates more iterations per second due to higher CFS scheduler weight. Both logs captured through the supervisor logging pipeline.

```
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ sudo ./engine logs lowpri
log: logs/lowpri.log
cpu_hog alive elapsed=1 accumulator=3210852331687054113
cpu_hog alive elapsed=2 accumulator=545427746515584643
cpu_hog alive elapsed=3 accumulator=4898054846622089754
cpu_hog alive elapsed=4 accumulator=13180482363536576754
cpu_hog alive elapsed=5 accumulator=10642260224213203843
cpu_hog alive elapsed=6 accumulator=12412610904322047729
cpu_hog alive elapsed=7 accumulator=8058724172362437278
cpu_hog alive elapsed=8 accumulator=2624195054828961553
cpu_hog alive elapsed=9 accumulator=17137611558208412396
cpu_hog done duration=10 accumulator=5905675852540279092
cpu_hog alive elapsed=1 accumulator=11803257113522441384
cpu_hog alive elapsed=2 accumulator=3482200311807596001
cpu_hog alive elapsed=3 accumulator=9405058507208106178
cpu_hog alive elapsed=4 accumulator=6901914695915585481
cpu_hog alive elapsed=5 accumulator=517941075213878912
cpu_hog alive elapsed=6 accumulator=6860804238671516944
cpu_hog alive elapsed=7 accumulator=727604665119969244
cpu_hog alive elapsed=8 accumulator=11346151119973165777
cpu_hog alive elapsed=9 accumulator=7808034488091734074
cpu_hog alive elapsed=10 accumulator=14141701957121838835
cpu_hog done duration=10 accumulator=14141701957121838835
cpu_hog alive elapsed=1 accumulator=15890837980296087904
cpu_hog alive elapsed=2 accumulator=5341742089692781985
cpu_hog alive elapsed=3 accumulator=1033759794101037729
cpu_hog alive elapsed=4 accumulator=17658878270256030959
cpu_hog alive elapsed=5 accumulator=3286759421113860782
cpu_hog alive elapsed=6 accumulator=13986909064823464989
cpu_hog alive elapsed=7 accumulator=7863634068817086463
cpu_hog alive elapsed=8 accumulator=4323336249010334567
cpu_hog alive elapsed=9 accumulator=18348486289131261944
cpu_hog alive elapsed=10 accumulator=7144446275009382094
cpu_hog done duration=10 accumulator=7144446275009382094
```

Screenshot 8 — Clean Teardown

Demonstrates: Supervisor shuts down cleanly, all containers reaped, logging thread joined, no zombies.

```
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ sudo ./engine ps
ID              PID     STATE     STARTED            LOG
lowpri          8164    exited    09:11:10           logs/lowpri.log
highpri         8159    exited    09:11:02           logs/highpri.log
lowpri          8152    exited    09:10:3
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ ps aux | grep -E "engine|cpu_hog|memory_hog" | grep -v grep
root        7995  0.0  0.1  17288  6084 pts/0    S+   08:59   0:00 sudo ./engine supervisor ./rootfs-base
root        7996  0.0  0.0  17288  2388 pts/2    Ss   08:59   0:00 sudo ./engine supervisor ./rootfs-base
root        7997  0.0  0.0  93604  1736 pts/2    Sl+  08:59   0:00 ./engine supervisor ./rootfs-base
```

```
sanchita@sanchita:~/projects/OS_Jackfruit/boilerplate$ sudo ./engine supervisor ./rootfs-base
[supervisor] Ready. rootfs=./rootfs-base socket=/tmp/mini_runtime.sock
^C[supervisor] Shutting down...
[supervisor] Exited cleanly.
```

[supervisor] Shutting down... then [supervisor] Exited cleanly. — all container processes stopped, bounded-buffer consumer thread joined, metadata freed, socket unlinked. No zombie processes remain.